Supplementary Materials

Supplementary Methods

Participating groups were asked to provide summaries of their prediction methods along with the actual predictions themselves. We provide these summaries (or, in some cases, modified versions) along with actual group names for those groups that agreed to do so, below:

Group 1 (Bologna Computing Group) – Submission 1. The submission is based on the prediction of stability changes in mutated proteins using INPS (Fariselli, et al., 2015) and INPS3D (Savojardo, et al., 2016) methods.

INPS (Impact of Non-synonymous mutations on Protein Stability) is a method for predicting the impact of non-synonymous Single Nucleotide Polymorphisms (nsSNPs) on protein stability, starting from protein primary sequence. INPS adopts a Support Vector Regression (SVR) approach, trained on seven features extracted from the protein primary sequence, including: BLOSUM substitution score, hydrophobicity (wild type and variants), Dayhoff mutability index of wild type, molecular weights of wild type and variant and evolutionary information derived from multiple sequence alignments.

In this study, we also used INPS3D, which extends INPS by also including, when available, features extracted from the protein 3D structure, namely: relative solvent accessibility and local energy change upon mutation (computed using pairwise residue contact potentials).

For the calibration procedure of the final scores, we used 11 known protein variants of TPMT reported in a functional characterization study by Salavaggione, et al. (2005). We used the distribution of INPS and INPS3D outcomes to fit a linear model used to remap raw stability change predictions onto the requested range (1=wild type, 0=totally destabilizing, >1 stabilizing). We assigned to each prediction value an arbitrary standard deviation of 0.1, that is quite low, to emphasize that we trust our method. The same calibration procedure was applied to both proteins.

For the non-sense variants, we adopt a different procedure based on the length of the variant. The prediction value corresponds to the ratio of the variant length over the WT length. Knowing that this approach is maybe too simplistic, we decided to assign to these predictions an arbitrary standard deviation of 0.2, that is greater than the standard deviation assigned to missense variants.

Group 1 (Bologna Computing Group) – Submission 2. The submission is based on the prediction of stability changes in mutated proteins using INPS (Fariselli, et al., 2015) and INPS3D (Savojardo, et al., 2016) methods.

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In this study, we also used INPS3D, which extends INPS by also including, when available, features extracted from the protein 3D structure, namely: relative solvent accessibility and local energy change upon mutation (computed using pairwise residue contact potentials).

For the calibration procedure of the final scores, we used 11 known protein variants of TPMT reported in a functional characterization study by Salavaggione, et al. (2005), and 27 known protein variants of PTEN collected from UniProt and from a functional characterization study by Lee, et al. (1999). We used the distribution of INPS and INPS3D outcomes to fit two linear model, one for each protein, used to remap raw stability change predictions onto the requested range (1=wild type, 0=totally destabilizing, >1 stabilizing). We assigned to each prediction value an arbitrary standard deviation of 0.1, that is quite low, to emphasize that we trust our method.

For the non-sense variants, we adopt a different procedure based on the length of the variant. The prediction value corresponds to the ratio of the variant length over the WT length. Knowing that this approach is maybe too simplistic, we decided to assign to these predictions an arbitrary standard deviation of 0.2, that is greater than the standard deviation assigned to missense variants.

Group 1 (Bologna Computing Group) – Submission 3. The submission is based on the prediction of stability changes in mutated proteins using INPS (Fariselli, et al., 2015) and INPS3D (Savojardo, et al., 2016) methods.

INPS (Impact of Non-synonymous mutations on Protein Stability) is a method for predicting the impact of non-synonymous Single Nucleotide Polymorphisms (nsSNPs) on protein stability, starting from protein primary sequence. INPS adopts a Support Vec-

tor Regression (SVR) approach, trained on seven features extracted from the protein primary sequence, including: BLOSUM substitution score, hydrophobicity (wild type and variants), Dayhoff mutability index of wild type, molecular weights of wild type and variant and evolutionary information derived from multiple sequence alignments.

In this study, we also used INPS3D, which extends INPS by also including, when available, features extracted from the protein 3D structure, namely: relative solvent accessibility and local energy change upon mutation (computed using pairwise residue contact potentials).

For the calibration procedure of the final scores, we used 11 known protein variants of TPMT reported in a functional characterization study by Salavaggione, et al. (2005), and 27 known protein variants of PTEN collected from UniProt and from a functional characterization study by Lee, et al. (1999). We used the distribution of INPS and INPS3D outcomes to fit two linear model, one for each protein, used to remap raw stability change predictions onto the requested range (1=wild type, 0=totally destabilizing, >1 stabilizing). We assigned to each prediction value an arbitrary standard deviation of 0.1, that is quite low, to emphasize that we trust our method.

For the non-sense variants, we adopt a different procedure based on the length of the variant. The prediction value corresponds to the ratio of the variant length over the WT length. Knowing that this approach is maybe too simplistic, we decided to assign to these predictions an arbitrary standard deviation of 0.2, that is greater than the standard deviation assigned to missense variants.

Finally we assigned a penalty to predictions of missense mutations that occur in positions known to be relevant for protein stability, as derived from UniProt and literature based observation (Wu, et al., 2007). The amount of the penalty is related to the chemical-physical similarity of the amino acid substitution, following the scores of the McLachlan matrix (McLachlan, 1971).

Group 2 – Submission 1. Benchmark datasets should sample the event space so that they could be used for method training and testing. Such datasets should fulfil several requirements. Relevance, i.e. that the cases represent the investigated feature, is one of them. A large number of methods have been developed to predict effects of amino acid substitutions on protein stability. Many of them utilize machine learning algorithms and have been trained with data from ProTherm database. When we examined cases in the database, we noticed a number of issues with the database contents and quality. Sever-

al issues emerged including some errors, but also features that had not been clearly communicated and which had caused data items to be used in wrong way by method developers.

We firstly checked thoroughly the details for variants in ProTherm database and corrected numerous problems. In the end, we had less than 50% of the original variants left. Out of these, 77% came from ProTherm, the rest are either corrected or new variants.

The final dataset contains 1564 entries from 99 proteins. This dataset is available in VariBench at http://structure.bmc.lu.se/VariBench/stability.php

With the new high-quality dataset we trained a novel machine learning predictor, PON-tstab, for amino acid substitution effects on stability and established a new baseline for variant stability prediction method performance. Our study revealed the importance of knowing and checking data that and their relevance when used for predictor development since predictors cannot be better than the data used to train them. Therefore, one has to be careful when using datasets collected by others unless they are properly documented and systematically compiled.

In total, 1106 features were collected to train a Random Forests based method for stability prediction. To eliminate redundant and non-relevant features, we used a combined greedy feature-selection algorithm with two steps: backward elimination and forward selection. Hence we got only on 8 features, including temperature, 1 conservation feature, 3 amino acid features, 2 neighbor features and 1 protein feature.

PON-tstab classifies the effect of variations on protein stability into three states: increase, decrease and no-change, also supplies a possibility_score (from 0 to 1) for the classification result.

The web service is available at http://structure.bmc.lu.se/PON-Tstab/. And the manuscript has been submitted to Journal of Molecular Biology.

To meet the format of CAGI challenge, we marked number 1 as Prediction result if the effect is predicted as “no change”, marked 1-possibility_score for “decrease” ones, marked 1+possibility_score for “increase” ones. We cannot predict wild type or mutation containing X, so these are marked as “*”.But for the “Standard_Deviation” column, we have no idea how to define the value for each mutation, just leave “*” for all mutations. We think each protein can has one SD value for all mutations’ prediction result, which is 0.608 for PTEN, and 0.63 for TPMT.

Group 3 – Submission 1. M47 is a SVM based predictor for the effects of Mutation on protein stability. To decrease the number of model parameters and to improve the generalization potential, we calculated amino acid contact energy (CE) change for point variations using a structure-based coarse-grained model. Based on the CE change and further physicochemical properties of amino acids as input features, we developed the support vector machine classifier with 47 input features.

To meet the format of CAGI challenge, we normalized Prediction result from 0-2. We cannot predict wild type or mutation containing X, so these are marked as "*".

For the "Standard_Deviation" column, we have no idea how to define the value for each mutation, just leave "*" for all mutations. We think each protein can has one SD value for all mutations' prediction result, which is 0.388 for PTEN, and 0.456 for TPMT.

The work is published (Yang, et al., 2013) and the tool can be downloaded from http://structure.bmc.lu.se/PPSC/

Group 4 – Submission 1. Protein stability is a consequence of the net balance of forces. Protein stability can be estimated from energy(G) (Worth, et al., 2011) and is also reflected in its flexibility(flexible regions present in a protein). While calculation of energy and its change is straight forward, it does not necessarily manifest directly in functional terms on biochemical activity and its alteration. Therefore, interpretation of flexibility graphs is an alternate paradigm one can explore for assessing single mutants for stability and functional consequences. In our case, the flexibility is calculated from the root mean square fluctuation of the structures in a Molecular Dynamics trajectory.

We ran the molecular dynamics on the C_α atoms of protein structure for 1 microsecond with CGMM (Coarse Grained Molecular Mechanics) force field. From the simulation, to obtain flexible regions we used the RMSF(Root Mean Square Fluctuations) values of all frames. These RMSF values are then normalized. These normalized RMSF values are real numbers so we convert them into a string as described in Bhadra and Pal (2014). On the basis of $RMSF_{norm}$ profile and the criterion for flexible region i.e. the percentage occurrence of a symbol (L > 35 or combined G,H and I > 14), we obtained the flexible regions of the structure. The symbols correspond to RMSF ranges of 0-1, 1-2, 2-3, > 3 for L, I, H, G, symbols, respectively. Since the number of mutations to investigate were very high, it was not feasible for us to run all the molecular dynamics simulations. For that we ran Multiple Sequence Alignment (MSA) of the structures from Clustalx (Jeanmougin, et al., 1998) with the neighbor joining method. We clustered them on

the basis of phylogenetic tree obtained from the MSA so that the number of simulation is computationally feasible. Then from the cluster we took one representative and did the molecular dynamics simulation on it. As the protein structures in one cluster are very similar so we estimated the flexibility of other mutant structures from this representative structure. Only in cases where the mutation was from/to Gly or Cys, we ran all the simulations, because the coarse-grained potential function corresponding to these two residues were showing significant variations in the fluctuations. But as we did not run the simulation on other proteins in the cluster we have given higher standard deviation for those proteins in the final results.

On obtaining the molecular dynamics trajectories, the normalized RMSF and its symbolic representation (Bhadra and Pal, 2014) was used corresponding to the wild-type and the mutant. A score was derived to infer the flexibility of the mutant and wild type to estimate closeness between the two structures. For this the weighted mean value was calculated from the frequency of symbols from the flexible regions weighted by the RMSF ranges they represent, yielding a score corresponding to global flexibility estimate corresponding to the protein. A difference of the count of the symbols between the two proteins in the common flexible regions (expressed through the weighted average as defined above), gave us an estimate if the protein became more rigid/flexible compared to the wild type. The values obtained were normalized between 0-2 with a mean at 1, corresponding to the wild type.

As described above, L, I, H, G corresponds to different RMSF ranges. We counted the occurrence of each symbol in the flexible regions of wild type and mutant. Then, a difference between these numbers for each letter was calculated. On the basis of these differences the score was calculated using the following formulae:

$$Score = \frac{l\times 1 + i\times 2 + h\times 3 + g\times 4}{l + i + h + g}$$

where, l = difference between occurrence of L letter in the flexible regions of wild type and mutant.

i = difference between occurrence of I letter in the flexible regions of wild type and mutant.

h = difference between occurrence of H letter in the flexible regions of wild type and mutant.

g = difference between occurrence of G letter in the flexible regions of wild type and mutant.

The global change in stability of the protein does not necessarily alter the function of the protein. There can be cases where the instability is in the distant region from the functionally important region hence not affecting the function of the protein; whereas, there can be cases where the protein is very much stable that it does not show the required amount of flexibility for proper functionality. So, in comment section we have written the distances between the ACV (auto-correlation vector) profiles of wild type and mutant flexible regions implying the difference in functionality i.e. the mutant shows the functional stability or not. This has important bearing on understanding the functional consequences of stability on the biochemical activity.

Group 5 (Lichtarge Lab) – Submissions 1 and 2. We predicted the effect of TPMT and PTEN variants on protein stability by using the Evolutionary Action (EA) method (Katsonis and Lichtarge, 2014).

The ***Evolutionary Action (EA)*** method estimates the fitness effect of each mutation and it does not use any training since it relies on a formal equation of the genotype-phenotype relationship. The terms of this equation were calculated using protein homology data. Briefly, the EA equation states that the fitness effect of a mutation equals the product of the sensitivity of the mutated position with the magnitude of the change. The sensitivity of the position is calculated by quantifying the correlation of the residue variations with phylogenetic branching within an alignment of homologous sequences (Lichtarge, et al., 1996; Lichtarge and Wilkins, 2010; Mihalek, et al., 2004). The magnitude of the change is calculated from substitution likelihood according to numerous sequence alignments for the given context (strata of sensitivity of the position, and optionally additional stratification based on structural features). The product is then normalized to represent the percentile rank of each variant within the protein in the scale of 0 (benign) to 100 (pathogenic). We used 87 homologous sequences to human ***PTEN*** (NP_000305) and 69 homologous sequences to human ***TPMT*** (NP_000358) as input, respectively. These sequences were selected to represent adequate evolutionary depth out of the hundreds of homologous sequences found by standard protein BLAST (Altschul, et al., 1997). The EA scores are available for all human variants at: http://mammoth.bcm.tmc.edu/EvolutionaryAction

The EA scores represent the fitness effect of each variant, which may be related to folding or to other interactions required for proper protein activity. Therefore, the protein stability effect (what the challenge asks for) is only one component of the fitness ef-

fect (what EA scores represent). To account for this difference and make proper prediction for this challenge, we assumed that the solvent accessibility of each protein residue can be used to de-couple the effect on folding (protein stability) from the effect on other interactions (effect beyond folding). Therefore, we calculated for each residue the fraction of its solvent inaccessible area, w_{fr} (we used the structures with PDB ID of 1d5r for PTEN and 2bzg for TPMT). To find whether this de-coupling helps, we made a second submission to serve as a control, where w_{fr} was set to 1 for all residues. The predicted values were calculated as: $1-w_{fr} \cdot EA/100$ (it matches the experimental value range of 0 (unstable) to 1 (wt stability). Silent variants were assumed to be stable (EA=0). Nonsense variants were annotated as unstable (EA=100), unless they occur beyond the folded domains (according to the aforementioned structures) and they were annotated as stable (EA=0).

Group 6 (Moult Lab) – Submissions 1 through 5. As a consequence of the short time line for this challenge and other activities within the group, we were able to spend only a very short time on this challenge. Thus, the approach is crude. Nevertheless, we decided to submit anyway, for the educational value.

We assumed that the experimental assay effectively measures protein abundance relative to wild type and that this quantity is closely related to the relative thermodynamic stability of proteins containing each mutation or to effects on specific biochemical mechanisms known to affect half-life such as ubiquitination.

We investigated the properties of three methods of estimating the effect of mutations on thermodynamic stability with structural data: SNPs3D stability (Yue, et al., 2005), Rosetta (Das and Baker, 2008), and FoldX (Guerois, et al., 2002). Rosetta and FoldX estimate DDG for a mutation. SNPs3D stability returns a yes/no binary estimate of whether DDG is likely greater or less than a threshold related to pathogenicity in monogenic disease, together with a confidence score. The threshold is approximately 3 Kcal/mol (Yue, et al., 2005).

We also used a sequence based SVR ensemble method (Yin, et al., 2017), developed in CAGI4, to estimate total activity for each mutation. The sequence based method is expected to provide an estimate of the effect on activity from all mechanisms, not just stability, and therefore to be noisy. The results were calibrated so that predictions for mutations known to have wild type or higher activities (taken from the literature and interspecies variants) had values close to 1.0, and mutations expected to have low

activity (cancer drivers) have values close to 0.1.

In other work (Yin and Moult, 2019) we have shown that a large fraction of the mutations in a protein core that decrease activity do so as a result of lowering stability. Thus, for these mutations sequence methods and stability methods should be in broad agreement. Comparison of the ensemble sequence method with both FoldX and Rosetta showed that for those mutations with a low ensemble predicted activity there is trend to large DDG, in agreement with expectations. But overall, there is little relationship between the sequence and these two structure methods. Nevertheless, we decided to include Rosetta based predictions, largely because of good performance in a previous CAGI challenge, mutations in P16 (Carraro, et al., 2017). Comparison of ensemble predicted activity and SNPs3D stability confidence scores showed an approximately linear (but noisy) relationship for predicted activities between zero and about 0.6. So we also based predictions on SNPs3D stability.

We also checked the literature for biochemical regulation of PTEN and TPMT stability (Leslie, et al., 2008; Rodriguez-Escudero, et al., 2011; Wu, et al., 2007) which assisted in deciding domain specific treatment. The PTEN ubiquitination information (Gupta and Leslie, 2016) helped to calibrate ensemble method scores.

For the ensemble method, the standard deviation for all mutations is set to the value derived in training (CAGI4). For Rosetta and SNPs3D, the standard deviations were estimated from the scatter in the comparison with the ensemble results.

The six submissions were compiled as follows:

Prediction 1: Based only on the Ensemble sequence method.

Prediction 2: Based only on the Rosetta method.

Prediction 3: Based on SNPs3D stability for mutations predicted to have an abundance between 0 and 0.6 of wild type, and on Ensemble for the rest.

Prediction 4: Based on Ensemble for core residues and on Rosetta for surface residues.

Prediction 5: For surface residues, based on SNPs3D stability for mutations predicted to have an abundance between 0 and 0.6 of wild type, and on Ensemble for the rest. For core residues, based on Ensemble.

Group 7 – Submissions 1 and 2. PDB structures of PTEN and TPMT are collected from PDB web site. There are five PTEN crystal structures: 1D5R, 5BUG, 5BZX, 5BZZ, 2KYL and 4O1V. 2KYL is a NMR structure with 10 models, contains coordinates from position 397 to position 403. We separated 2KYL structure into 10 structures. 4O1V is a

hetero-dimer, of which chain B contains the coordinates from position 357 to position 363. All other structures contain coordinates from position 14 to position 351. There are two TPMT structures: 2BZG and 2H11. 2H11 is a homo-dimer with two TPMT chains. All TPMT structures contain coordinates from position 17 to position 245.

In order to predict the effects of missense mutations on structural stability of PTEN and TPMT proteins, we used two popular methods to predict the change of protein stability (DDG): Foldx (Schymkowitz, et al., 2005) and Rosetta (Kellogg, et al., 2011). DDG of a monomeric protein induced by a point mutation is defined as the difference in free energy between the mutant structure and the wildtype structure (e.g. DDG = mutant energy – wildtype energy). The negative DDG values indicate increased stability. The effect of mutations was calculated from these DDGs, which is described as following.

Method 1: PositionScan of FoldX

The command PositionScan of Foldx was used to mutate each selected position to target amino acids on the monomer chains of all structures. The protein position numbers were converted into PDB residue numbers by SIFTS (Velankar, et al., 2013). After DDG values were parsed from Foldx outputs, PDB residue numbers were converted back to protein position numbers.

The great majority of the mutations lied between -3 and 6 kcal/mol, which is consistent to the study of Faure and Koonin (2015). Given that the free energy of protein folding is distributed roughly between 5 and 15 kcal/mol Sikosek and Chan (2014), the effect values of mutations were calculated from prediction DDGs by the function:

$$Effect\ values = \begin{cases} 0, & DDG > 5 \\ \dfrac{(5 - DDG)}{5}, & DDG \leq 5 \end{cases}$$

The mean and standard deviation were calculated from the converted effect values over 13 monomer structures of PTEN protein and 3 monomers of TPMT protein. For those residues without coordinates, IUPred (Dosztanyi, et al., 2005) was used to predict their disorder probabilities. Those residues were predicted to be disordered or in a loop, N-terminus and C-terminus of the structures. We filled in 1.0 as effect values and 0.001 as standard deviations.

Method 2 : ddg_monomer application in the program of Rosetta

We used the High Resolution Protocol of ddg_monomer application (https://www.rosettacommons.org/docs/latest/application_documentation/analysis/ddg-monomer). First, the structures were pre-minimized by mini-

mize_with_cst.linuxgccrelease function, output a minimization log file for each structure. Second, a distance restraint file was generated from the minimization log file by convert_to_cst_file.sh. Third, PTEN and TPMT mutation data sets were converted into Resfile format input. The parameters of ddg_monomer were set to default settings. The number of iterations was set to 10. Fourth, we used five different methods to obtain energy scores of wildtype and mutants: (1). the average energy of 10 decoys; (2). the minimum energy; (3). the median energy. (4). the average of top-3-scoring decoys. (5). the third top scoring decoys. A DDG was calculated by subtracting wildtype energy from mutant energy. We used all five DDGs for each mutant and each structure. Last, we used the same function in "Foldx Method" to calculate the mutation effect values from DDGs. For a position, if there are 20 prediction values, we removed the bottom 5 and top 5, calculated the average and standard deviation from the middle 10 values. If there is only one structure, all five prediction values were used.

Group 8 (Zhou Lab) – Submission 1. All predictions were made using EASE-MM: Evolutionary, Amino acid, and Structural Encodings with Multiple Models (Folkman, et al., 2016), a method developed previously and available as a web-server at http://sparks-lab.org/server/ease/. The method predicts protein stability changes with a protein sequence and mutation as the only inputs, without the three-dimensional structure of the protein. Instead, structural properties of the protein are predicted using SPIDER2 (Heffernan, et al., 2015). EASE-MM comprises five specialized support vector regression (SVR) models to predict $\Delta\Delta G_u$ of mutations in residues located in different secondary structure (SS) elements (helix, sheet, or coil) and with different levels of accessible surface area (ASA) (exposed or buried with a 25% threshold). The final prediction is the average of $\Delta\Delta G_u$ predicted with two models, one selected based on the predicted SS and the other based on the predicted ASA of the mutation site. We used a dataset of 1676 mutations (70 proteins) from the ProTherm database (Kumar, et al., 2006), version February 2013, to design the method and estimate its performance using 10-fold cross-validation (CV), and a dataset of 236 mutations (23 proteins) with a low sequence identity (<25%) to the design dataset in order to independently test our method. Importantly, we verified all records in ProTherm and corrected incorrect entries according to the original publications.

To build the five models employed by EASE-MM, we partitioned the design dataset according to SS and ASA predicted from the protein sequence (SPIDER2 correctly

predicted the SS (ASA) of 80% (84%) and 81% (86%) of PTEN and TPMT residues, respectively). A unique set of predictive features was identified for each of the five SVR models using the sequential forward floating search (Pudil, et al., 1994). As a result, each model included a unique combination of evolutionary conservation features (such as the difference of the wild-type and mutant amino acid probabilities in a multiple sequence alignment), amino acid parameters (such as differences in volume, bulkiness, hydrophobicity of the wild-type and mutant residues) and predicted structural properties (such as SS elements and ASA). Importantly, these features were selected using "unseen-protein" 10-fold CV, which was devised to avoid over-fitting on specific proteins by splitting the dataset into CV folds so that all mutations of a cluster of similar proteins (≥ 25% sequence identity) are always contained within a single fold (Folkman, et al., 2014). The same CV scheme was employed to optimize hyper-parameters of the SVR models and the radial basis function kernel (C, γ, and ε) using grid search. Finally, to maximize the training dataset size, we merged the design and test datasets before training the final models which were used in this challenge and are available on our web-server. Of note, no PTEN and TPMT mutations or mutations of related proteins (≥ 25% sequence identity) were included in the training dataset.

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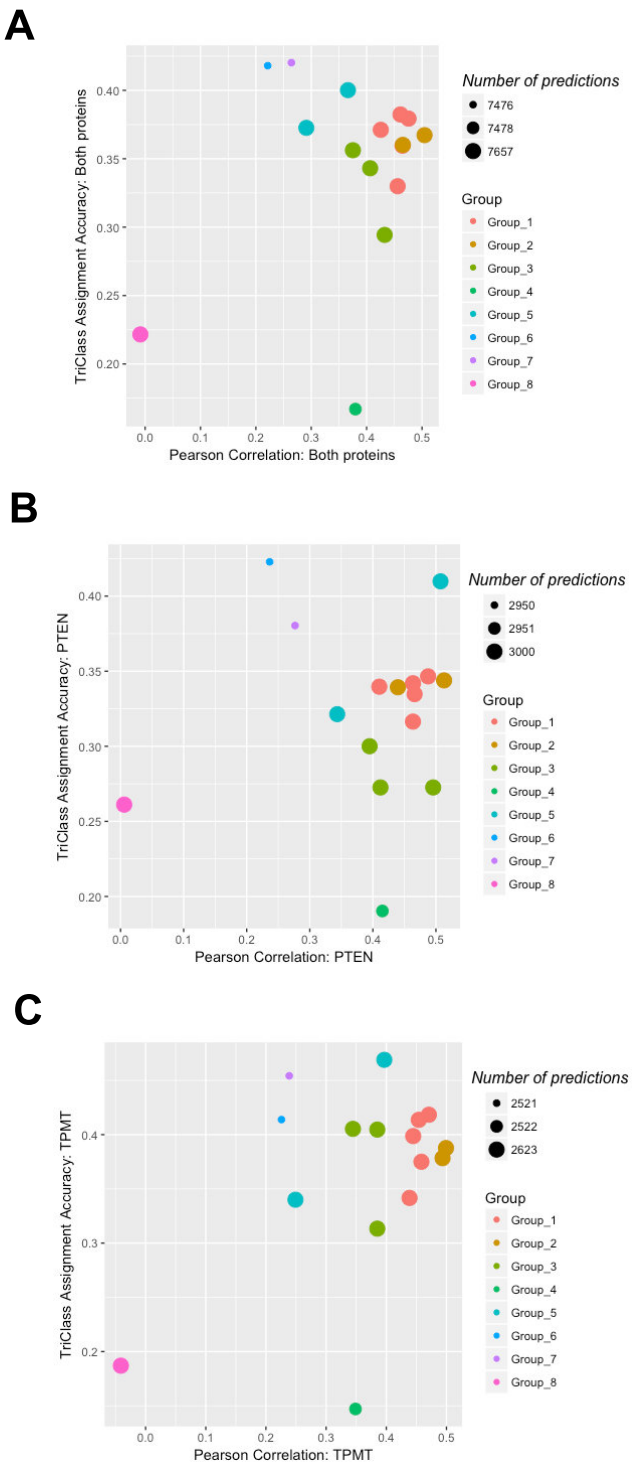
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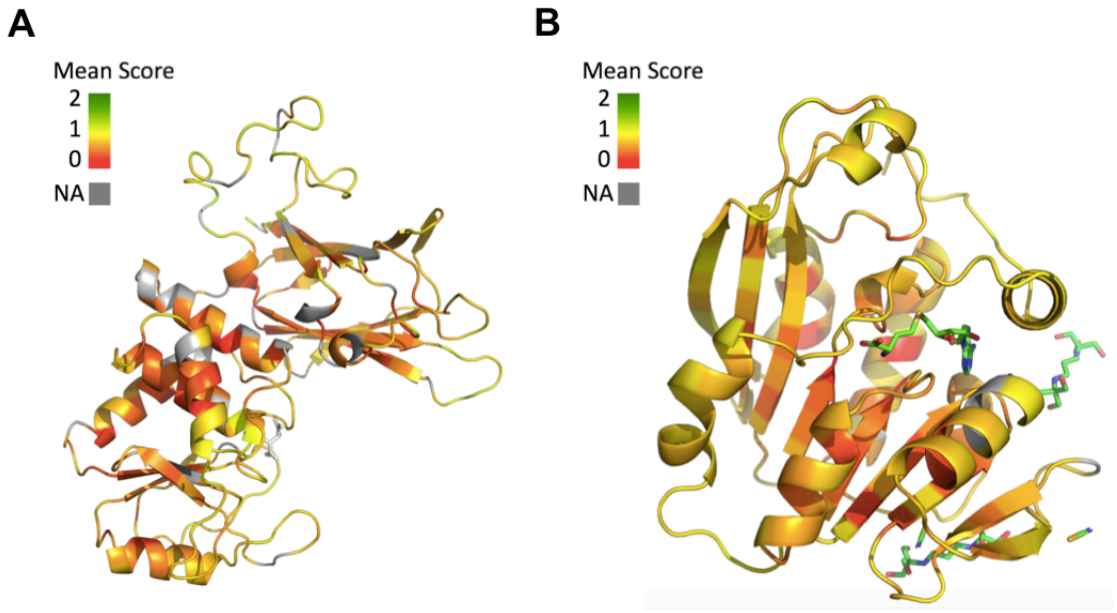
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Supplementary Figures



Supplementary Figure S1. Performance as a function of number of predictions made by each submitted method when considering **(A)** both proteins, **(B)** PTEN only and **(C)** TPMT only.



Supplementary Figure S2. Spatial distribution of VSP stability scores in the crystal structures of **(A)** PTEN, and **(B)** TPMT. PDB IDs for the crystal structures for PTEN and TPMT: 1D5R and 2H11:A, respectively. For each position, the mean score is shown.

Supplementary Tables

Supplementary Table S1. Summary of predictor data sets and performance measures when variants with negative stability scores were set to zero. Bold values represent the best-performing submission for a given performance measure. When the best-performing method was a baseline method, its value is shown in red font, and the values for the top submitted method are bold.
See spreadsheet

Supplementary Table S2. Summary of predictor data sets and performance measures when variants with negative stability scores were excluded. Bold values represent the best-performing submission for a given performance measure. When the best-performing method was a baseline method, its value is shown in red font, and the values for the top submitted method are bold.
See spreadsheet